



More Factoring Patterns

Polynomials are written in many different forms, some of which are recognizable factoring patterns.

Factoring and Checking the Sum and Difference of Cube Patterns

You can check whether your factors are correct by multiplying them together.

EXAMPLE 1

Factor and use multiplication to check the answer.

A $27x^3 - 64$

SOLUTION

The binomial $27x^3 - 64$ can be written as $(3x)^3 - 4^3$ and can be factored with the difference of cubes pattern.

$$27x^3 - 64 = (3x - 4)(9x^2 + 12x + 16)$$

CHECK

$$\begin{array}{r} 9x^2 + 12x + 16 \\ \times 3x - 4 \\ \hline -36x^2 - 48x - 64 \\ 27x^3 + 36x^2 + 48x \\ \hline 27x^3 - 64 \quad \checkmark \end{array}$$

B $8x^3 + 125$

SOLUTION

The binomial $8x^3 + 125$ can be written as $(2x)^3 + 5^3$ and can be factored with the sum of cubes pattern.

$$8x^3 + 125 = (2x + 5)(4x^2 - 10x + 25)$$

CHECK

$$\begin{array}{r} 4x^2 - 10x + 25 \\ \times \quad \quad \quad 2x + 5 \\ \hline 8x^3 - 20x^2 + 50x \\ 20x^2 - 50x + 125 \\ \hline 8x^3 \qquad \qquad \qquad + 125 \end{array} \quad \checkmark \quad \blacksquare$$

Factoring by Grouping

Some polynomials can be factored by grouping. With this method, a common polynomial term is factored from pairs of terms.

EXAMPLE 2

Factor.

A $x^3 + 2x^2 - 25x - 50$

SOLUTION

Group the polynomial by terms and factor out a common binomial from each group.

$$\begin{aligned} x^3 + 2x^2 - 25x - 50 &= (x^3 + 2x^2) - (25x + 50) && \text{Group into pairs of terms.} \\ &= x^2(x + 2) - 25(x + 2) && \text{Factor the common binomial.} \\ &= (x^2 - 25)(x + 2) && \text{Distributive Property} \\ &= (x - 5)(x + 5)(x + 2) && \text{Factor the difference of squares.} \end{aligned}$$

B $2y^2 + 5y - 7$

SOLUTION

It doesn't look like you can group this polynomial, but if you can find a way to split the middle term, you will be able to group the polynomial by its terms.

$2 \cdot (-7) = -14$	Find the product ac .
$-2, 7$	Find two factors of -14 whose sum is 5.
$2y^2 - 2y + 7y - 7$	Rewrite $5y$ as the sum of $-2y$ and $7y$.
$= (2y^2 - 2y) + (7y - 7)$	Group pairs of terms.
$= 2y(y - 1) + 7(y - 1)$	Find a common binomial factor for each group.
$= (2y + 7)(y - 1)$	Distributive Property ■

Factoring Completely

To factor a polynomial completely, first look for a greatest common factor. If you find one, factor it out of the polynomial. Examine what remains. Look for special factors (difference of squares, difference of cubes, sum of cubes, perfect square trinomial) and try to factor by grouping. You can use the guess-and-check strategy to factor trinomials of the form $x^2 + bx + c$.

► **Remember** If a polynomial cannot be factored, it is called prime.

EXAMPLE 3

Factor completely.

A $4x^4 - 4$

SOLUTION

$4x^4 - 4 = 4(x^4 - 1)$	Factor out the greatest common monomial factor.
$= 4(x^2 + 1)(x^2 - 1)$	Factor the difference of squares.
$= 4(x^2 + 1)(x + 1)(x - 1)$	Factor another difference of squares.

B $-32z^2 - 48z - 18$

SOLUTION

$$-32z^2 - 48z - 18 = -2(16z^2 + 24z + 9)$$

Factor out the greatest common monomial factor.

$$= -2(4z + 3)^2$$

Factor the perfect square trinomial. ■

EXAMPLE 4

Factor completely. If the polynomial is not factorable, write *prime*.

A $20x^4 + 80x^3 - 5x^2 - 20x$

SOLUTION

$$20x^4 + 80x^3 - 5x^2 - 20x$$

$$= 5x(4x^3 + 16x^2 - x - 4)$$

Factor the greatest common monomial factor.

$$= 5x[(4x^3 + 16x^2) - (x + 4)]$$

Group into pairs of terms.

$$= 5x[4x^2(x + 4) - (x + 4)]$$

Find a common binomial factor for each group.

$$= 5x(4x^2 - 1)(x + 4)$$

Distributive Property

$$= 5x(2x + 1)(2x - 1)(x + 4)$$

Factor the difference of squares.

B $25x^4 + 16y^6$

SOLUTION

The binomial $25x^4 + 16y^6$ is prime because it has no common monomial factors and is the sum of two squares, $(5x^2)^2$ and $(4y^3)^2$.

C $40a^3 + 135b^6$

SOLUTION

$$40a^3 + 135b^6 = 5(8a^3 + 27b^6)$$

Factor the greatest common monomial factor.

$$= 5[(2a)^3 + (3b^2)^3]$$

Rewrite the factor as the sum of cubes.

$$= 5[(2a + 3b^2)((2a)^2 - 2a \cdot 3b^2 + (3b^2)^2)]$$

Factor the sum of cubes.

$$= 5(2a + 3b^2)(4a^2 - 6ab^2 + 9b^4)$$

Simplify.

D $300 - 3(x - 1)^2$

SOLUTION

$$300 - 3(x - 1)^2 = 3 \left[100 - (x - 1)^2 \right]$$

$$= 3 \left[\mathbf{10} - (\mathbf{x} - \mathbf{1}) \right] \left[\mathbf{10} + (\mathbf{x} - \mathbf{1}) \right]$$

$$= 3(10 - x + 1)(10 + x - 1)$$

$$= 3(-x + 11)(x + 9)$$

Factor the greatest common monomial factor.

Factor the difference of two squares.

Simplify.

Simplify.

E $1 - t^6$

SOLUTION

$$1 - t^6 = (1 + t^3)(1 - t^3)$$

$$= (1 + t)(1 - 2t + t^2)(1 - t)(1 + 2t + t^2)$$

Factor the difference of squares.

Factor the sum of cubes and the difference of cubes. ■